landuse.lum

Purpose

`landuse.lum` is the master table of land use and management classes. Each row defines one class by name and lists string pointers to records in other files: the plant community (`plants.ini`), management schedule (`management.sch`), curve number entry (`cntable.lum`), conservation practice (`cons_practice.lum`), urban land use (`urban.urb`), overland-flow Manning's n class (`ovn_table.lum`), tile drainage (`tiledrain.str`), septic (`septic.str`), filter strip (`filterstrip.str`), grassed waterway (`grassedww.str`), and user BMP (`bmpuser.str`). HRUs reference a row in this table through `hru-data.hru`. After reading, the reader resolves every non-`null` string pointer to an integer index stored in `lum_str`; unresolved names are logged to `simulation.out` (unit 9001).

Source

- Reader: src/landuse_read.f90
- Type definition: src/landuse_data_module.f90, `type land_use_management` (instances stored in `lum`)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- One record per land use class. Records are list-directed (free format). All 14 fields are read on every row.
- Any pointer field that should be empty must be the literal string `null`. The reader treats `null` as "no link" and does not try to resolve it.

#	Field	Type	Default	Description
1	name	char(40)	blank	name of the land use and management class. Referenced from `hru-data.hru`
2	cal_group	char(40)	blank	calibration group label. Not currently used by the solver
3	plant_cov	char(40)	blank	plant community pointer into `plants.ini`. Resolved to `lum_str%plant_cov`
4	mgt_ops	char(40)	blank	management schedule pointer into `management.sch`. Resolved to `lum_str%mgt_ops`
5	cn_lu	char(40)	blank	curve number entry pointer into `cntable.lum`. Resolved to `lum_str%cn_lu`
6	cons_prac	char(40)	blank	conservation practice pointer into `cons_practice.lum`. Resolved to `lum_str%cons_prac`
7	urb_lu	char(40)	blank	urban land use pointer into `urban.urb` (e.g. `URLD`, `URHD`). Resolved during urban init
8	urb_ro	char(40)	blank	urban runoff model. `usgs_reg` for USGS regression equations, `buildup_washoff` for build-up/wash-off
9	ovn	char(40)	blank	overland-flow Manning's n class pointer into `ovn_table.lum`
10	tiledrain	char(40)	blank	tile drainage pointer into `tiledrain.str`. Resolved to `lum_str%tiledrain`
11	septic	char(40)	blank	septic pointer into `septic.str`. Resolved to `lum_str%septic`

#	Field	Type	Default	Description
12	fstrip	char(40)	blank	filter strip pointer into `filterstrip.str`. Resolved to `lum_str%fstrip`
13	grassww	char(40)	blank	grassed waterway pointer into `grassedww.str`. Resolved to `lum_str%grassww`
14	bmpuser	char(40)	blank	user-defined BMP pointer into `bmpuser.str`. Resolved to `lum_str%bmpuser`

The reader logs a line to `simulation.out` for every non-`null` pointer that fails to match a name in the target file. The simulation continues with the link unresolved (index zero), which means the corresponding behaviour is skipped.

Example

Excerpt from `refdata/Ames_sub1/landuse.lum`:

```
landuse.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                          cal_group          plnt_com                                      mgt
cosy_lum                           null         cosy_comm                                   mgt_01
agrr_lum                           null         agrr_comm                                 AGRR_rot
frst_lum                           null         frst_comm                                     null
urld_lum                           null              null                                     null
```

Notes:

- The header tokens written by SWAT+ Editor (`plnt_com`, `cn2`, `urban`, `ov_mann`, `tile`, `sep`, `vfs`, `grww`, `bmp`) do not match the field names in the source (`plant_cov`, `cn_lu`, `urb_lu`, `ovn`, `tiledrain`, `septic`, `fstrip`, `grassww`, `bmpuser`). The reader skips the header and uses position only.
- `urld_lum` is an urban class. Its `plant_cov` and `mgt_ops` are both `null`; the row links instead to `URLD` in `urban.urb` and selects `buildup_washoff` as the urban runoff model.
- `frst_lum` has no management schedule (`mgt_ops = null`). Forest classes typically rely on the plant community initialization alone.

Related files

- `hru-data.hru`. Each HRU points to one `landuse.lum` row by name.
- `cntable.lum`. Curve number table referenced by the `cn_lu` field. Each row holds a name and four curve numbers (hydrologic conditions A through D).
- `cons_practice.lum`. Conservation practice table referenced by `cons_prac`. Holds USLE P factor and maximum slope length per practice.
- `ovn_table.lum`. Overland-flow Manning's n table referenced by `ovn`. Holds mean, minimum, and maximum n per class.
- `plants.ini`. Target of `plant_cov`.
- `management.sch`. Target of `mgt_ops`.
- `urban.urb`. Target of `urb_lu`.
- `tiledrain.str`, `septic.str`, `filterstrip.str`, `grassedww.str`, `bmpuser.str`. Structural BMP files targeted by the last five pointer columns.